

6.4.10 Software Maintenance Process

This International Standard specializes the Software Maintenance Process (aligned with the Operation & Maintenance concepts in ISO/IEC 15288). Organizations may claim conformance to the relevant 15288 processes or this subclause (6.4.10).

6.4.10.1 Purpose

The purpose of the Software Maintenance Process is to provide cost-effective support for a delivered software product throughout its operational life. Pre-delivery activities include planning for post-delivery operations and supportability; post-delivery activities include software modification and operational support (e.g., training, logistics, and service continuity).

6.4.10.2 Outcomes

As a result of successful implementation of the Software Maintenance Process:

- a) A maintenance strategy is defined for managing product modification, migration, and release cadence.
- b) Impacts of changes on the organization, operations, security, safety, data/privacy, and interfaces are identified and evaluated.
- c) Affected system/software documentation, configuration baselines, and records are updated.
- d) Modified products are developed with associated verification/validation and regression tests to ensure requirements and non-functional constraints are not compromised.
- e) Product upgrades are migrated/deployed into the customer's environment with defined rollback and recovery strategies.
- f) Modifications, deprecations, and end-of-support notices are communicated to all affected parties.
- g) Maintenance records (requests, analyses, approvals, test evidence, release notes) are captured and retained per policy.

6.4.10.3 Activities and tasks

The maintainer shall implement the following activities in accordance with applicable organizational policies and procedures.

6.4.10.3.1 Maintenance planning and process implementation

- 6.4.10.3.1.1 Develop, document, and execute plans and procedures for conducting maintenance activities and tasks, including SLAs/SLOs, triage policy, classification, environments, and tooling.
- 6.4.10.3.1.2 Establish or integrate with the **Configuration Management Process** (6.3.4) for managing modifications to baselined items (code, data,

infrastructure, docs).

- 6.4.10.3.1.3 Define criteria for acceptance, regression scope, rollback/recovery, and release documentation (release notes, advisories).
- 6.4.10.3.1.4 Define interfaces with **Problem Resolution** (7.2.8), **Verification** (7.2.4), **Validation** (7.2.5), **Review** (7.2.6), **Audit** (7.2.7), **Installation** (6.4.7), and **Operation** (6.4.9).

6.4.10.3.2 Problem and modification analysis

- 6.4.10.3.2.1 Analyze the impact of problem reports or modification requests on the organization, existing system, and interfacing systems. Classify by:
 - a) **Type**: corrective, improvement, preventive, adaptive.
 - b) **Scope**: size/complexity, cost, schedule.
 - c) **Criticality**: performance, safety, security, compliance, availability.
 - 6.4.10.3.2.2 Reproduce or verify the reported problem.
 - 6.4.10.3.2.3 Develop options for implementing the modification (including do-nothing, workaround, phased fix, full remediation).
 - 6.4.10.3.2.4 Document the request, analysis, options, recommended option, and assumptions; link to CM items and test impact.
 - 6.4.10.3.2.5 Obtain approval for the selected option per contract or **Decision Management** (6.3.3).
 - 6.4.10.3.2.6 Register and track the item in **Problem Resolution** (7.2.8).

6.4.10.3.3 Modification implementation

- 6.4.10.3.3.1 Determine which documentation, software units, data schemas, configurations, and versions require change; update traceability.
- 6.4.10.3.3.2 Enter the **Technical Processes** (6.4) to implement the change; define test & evaluation criteria for modified and unaffected areas (regression).
- 6.4.10.3.3.3 Perform development, peer reviews, static/dynamic analysis, and security checks; maintain branch/merge discipline under CM.
- 6.4.10.3.3.4 Execute verification (7.2.4) and, where appropriate, validation (7.2.5); ensure original requirements remain satisfied.
- 6.4.10.3.3.5 Record test results and evidence; update user/admin docs, runbooks, and training materials.

6.4.10.3.4 Maintenance review and acceptance

- 6.4.10.3.4.1 Conduct reviews with the authorizing organization to assess integrity, completeness, and risks of the modified system (see 7.2.6).
- 6.4.10.3.4.2 Support audits as required (7.2.7).
- 6.4.10.3.4.3 Obtain approval/sign-off for satisfactory completion per contract/quality plan.

6.4.10.3.5 Migration

- 6.4.10.3.5.1 Ensure software and data produced/modified during migration comply with this International Standard and contractual requirements.
- 6.4.10.3.5.2 Develop, document, and execute a migration plan with user involvement, covering requirements analysis, tooling, data/model conversion, dry-runs, execution, verification/validation, back-out/rollback, and future support for the old environment.
- 6.4.10.3.5.3 Notify users of migration plans, including rationale for ending support, details of the new environment, timelines, and support options.
- 6.4.10.3.5.4 Execute changeover/cutover or concurrent operation as planned; maintain service continuity and quality; coordinate with **Operation** (6.4.9).
- 6.4.10.3.5.5 Announce commencement of the scheduled migration; upon completion, archive old-environment documentation, logs, and code per retention policy.
- 6.4.10.3.5.6 Perform a post-operation review to assess migration outcomes; report results to the appropriate authorities.
- 6.4.10.3.5.7 Ensure data from the old environment remains accessible per contract and applicable data protection/audit requirements.

6.4.10.3.6 Release to operation and support handover

- 6.4.10.3.6.1 Package the release (artifacts, configs, infrastructure changes, release notes, advisories) and coordinate **Installation** (6.4.7).
- 6.4.10.3.6.2 Handover to **Operation** (6.4.9): update runbooks, monitoring thresholds/SLOs, on-call procedures, and training.
- 6.4.10.3.6.3 Communicate deprecations and end-of-support timelines; provide workarounds when applicable.

6.4.10.3.7 Records and communication

- 6.4.10.3.7.1 Maintain maintenance records: requests, analyses, approvals, test evidence, reviews/audits, release notes, deprecation/EOS notices.
- 6.4.10.3.7.2 Communicate changes and impacts to affected stakeholders and partners; update stakeholder-facing documentation and portals.

Cross-references

- **Problem Resolution** (7.2.8) — log, track, and close issues.
- **Configuration Management** (6.3.4) — control of baselined items.
- **Verification** (7.2.4), **Validation** (7.2.5), **Review** (7.2.6), **Audit** (7.2.7) — assurance activities.
- **Installation** (6.4.7) and **Operation** (6.4.9) — deployment and service continuity.
- **Measurement** (6.3.2) — maintenance KPIs (MTTR, change fail rate, rollback rate, availability).